A Mixed-Effects re-analysis of TMNRED EEG dataset: disentangling linguistic features effects

TMNRED is an open access eeg dataset recorded while chinese subjects read 400 chinese sentences (target and nontarget) divided in 8 blocks. In the current study eeg features were computed on MNE python, alongside linguistic features of such sentences computed through natural language processing and methodological features such as repetitions of the sentences across blocks and target-nontarget group division based on the original research design. A new approach based on Linear Mixed Modelling was used in order to test whether fixed effects would differ from the original study when taking into account repetitions, linguistic features, subject and sentence variability. RSA was used to test the relationship between representational dissimilarity matrices of neural responses to such sentences and a semantic representational similarity matrix between the same sentences. Results showed: smaller group effects and new significant linguistic features effects, with between-subject and trial-level variability dominating fixed effects; non significant RSA similarity between neural pattern and semantic similarity across sentences; classifiers accuracy in binary classification (target vs nontarget) mainly accounted for by linguistic features.

BACKGROUND

Language is one of the main advanced cognitive human functions. One of the core processes of language cognition is semantic semantic processing, which functions as the foundation for human thought and behavior. Neuroimaging techniques allow extraction and analysis of neural activities underlying semantic encoding [1].

Brain rhythms are surely one of the possible measures of brain activity which are used for this purpose.

Regarding processing of sentences syntactic binding is possibly the main process underlying such function, that is the combination of words into larger structures. Brain rhythms seem to play a core role in syntactic binding, as older adults in older adults syntactic binding has been associated with increases in theta, alpha and beta power. In particular, modulations in the alpha and beta band are thought to underlie the actual binding taking place [7]. Specifically, semantic (retrieval) processes and any other highly cognitive loaded processes,

such as syntactic dependencies processing, have been associated with modulation of alpha and beta power to support storage of information [7,8].

The current study is based on an open access EEG dataset, namely “TMNRED”, for semantic target identification in natural reading environments. The dataset was obtained from 30 participants who read 400 divided in 8 blocks, one for each of the 8 recording sessions. The sentences were derived from public internet and media sources, and divided into two groups: target semantic materials and non-target semantic materials, where target materials were of four types of semantics (i.e. names, means of transportation, animals, and fruits). The language used in such setting was Chinese, which, contrary to English, is a logographic language. Moreover, as an ideographic writing system, each character contains information on form, sound and semantics, with extensive presence of polysemous words and homophones [1]. Its word order rules and syntactic structures pose higher demands on the linguistic cognitive processes. The authors of such dataset originally tested for a group effect on eeg recording of channels of interest (Cz, C3, Pz, Oz) and of PSD features.
The present study aims at testing these results on the relationship between eeg features and sentences semantic features, while taking into account variability within the experimental design, namely: sentence repetition across blocks, linguistic features of each sentence (i.e. word and character frequencies, number of strokes, surprisal, syntactic complexity), random subject and sentence variability. Moreover an ulterior objective was to carry out a Representational Similarity Analysis to further test whether the eeg responses to such sentences would be significantly related to their semantic similarities. Finally, classifiers would be used to assess the unique and combined contribution of nlp features and eeg features in distinguishing between target and nontarget sentences. The initial approach involved using across subject cross-validation only, but following LMM results, a pipeline for within subject cross validation was implemented as well with the intention of testing whether performances would differ when controlling for between subjects variability. Expected results included a weaker group effect when taking into account the above confounding variables and variability, and weak similarity from RSA, with weak performance of eeg based classifier and features not significantly improving the accuracy of eeg-nlp based classifiers compared to the nlp based classifier. As for the comparison between across-subjects and within subject models, better performances in within-subject midels were expected due to between-subject variability control.

METHODS

An MNE python pipeline was used in order to carry out preprocessing of raw eeg data front he TMNRED dataset [1].The pipeline followed the steps carried out in the dataset paper and included: bad channel detection with channels with power spectra greater than twice the standard deviation of the mean power spectrum marked as bad; interpolation; re-referencing; 0.5-80Hz bandpass filter and 50Hz notch filter; downsampling to 200Hz; ICA; epoching based on the duration of each trial. Trials with median variance of each channel twice the standard deviation and median difference between each channel and its mean twice the standard deviation were dropped (4,5%).

An NLP pipeline was implemented in Python in order to extract linguistic and semantic features for each sentence. Syntactic complexity was computed as the maximum depth of the dependency tree derived from Stanza’s dependency parse. Lexical surprisal was

computed using the GPT-2 language model as the negative log-probability of each token given its preceding context, then sentence-level surprisal was computed as the average of word-level values [3,6]. The number of strokes of each sentence was computed as the mean of the number of strokes of all Chinese characters of the sentence, which were obtained from a stroke-count dictionary derived from the Unihan database [4]. Lexical frequency of each sentence was estimated using SUBTLEX-CH norms, taking both the mean log character frequency and the mean log word frequency [2,4]. In addition, sentence-level semantic representations were obtained using Sentence-BERT embeddings, and cosine similarity was computed for all possible pairs of the 400 sentences in order to generate a semantic rappresentational similarity matrix [5].

The dataset derived from these pipelines included: subjects' ids; sessions' ids corresponding to the id of the block of sentences; group (i.e. target or nontarget sentence); unique labels of each sentences, that is when the sentence is repeated the label remains the same both times; labels indicating whether it is the first or second occurrence of the sentence; NLP features, namely syntactic complexity, surprisal, strokes, char_frequency, word_frequency; early (0-500ms) and late (>500ms) EEG features, namely amplitude of channel of interest, amplitude of ROI (i.e. clusters of channels), theta-alpha-beta-gamma power in channel of interest, theta-alpha-beta-gamma power in ROI (e.g. left beta activation); Z_scores of the above features.
Considering the numerosity of the dataset (N=11455) and the nature of my research questions, z-scores were used to standardize the NLP variables and the EEG variables, in particular regarding the variables related to PSD features a natural logarithmic function was applied before the computation of their z-scores.
From literature [4] it emerged that linguistic features can sometimes show multicollinearity, therefore a correlation matrix was used to assess whether the NLP variables were highly correlated.
The original dataset splits 400 sentences in 8 blocks of 50 each, where some sentences repeat themselves across blocks. In order to assess the possible impact of such repetition on the statistical analysis a contingency table with χ^2 test was carried out.
An independent samples T-Test with all the nlp features as dependent variables, and the grouping variable "Group" (i.e. target-nontarget sentences) was used to test whether the means of the nlp features were different among groups (target-nontarget). The dataset derived from the eeg preprocessing is characterized by dropped trials, therefore the dataset including all 400 sentences corrected by their repetition among blocks was modified by deleting the repeated sentences (from 400 sentences the dataset narrowed down to 276), before the t-test.
In order to replicate the group effect of the original TMNRED paper a paired t-test was carried out on a dataset where each line representing a subject was built, instead of representing one sentence. Using such dataset, where each line represents a single trial, Linear Mixed Models were carried out to test whether there was a Group effect on language processing-related eeg features as cited in the original paper [1], while controlling for both repetitions and nlp features.

Representational Dissimilarity Matrices (RDM) were computed for each subject, taking into consideration only the eeg features related to semantic processing in their late components (i.e. (≥ 500ms) [1], namely Cz, C3, Pz, alpha late global activation and beta late left activation (computed by averaging across F3, C3, P3 beta activation). For each subject,

RDMs were computed using correlation distance (1 − Pearson correlation) only across unique trials, that is after deleting all repetitions of the same sentence and keeping only the first occurrence of it. This approach was chosen over averaging across repetitions after testing for spearman correlation between RDM based on the first occurrence only, and RDM based on the second occurrence only.
For each subject, Representational Similarity Analysis (RSA) was performed by correlating the upper triangle of the neural RDM with the corresponding semantic dissimilarity matrix using Spearman correlation. All subjects results were averaged together in order to obtain a group-level result.
A permutation-based sign-flipping test (10,000 permutations) was used to assess whether the group-level mean RSA differed significantly from zero.

Supervised binary classifiers, using Logistic regression, were used in order to assess whether the available eeg and nlp features contained sufficient information to reliably distinguish target from nontarget sentences. Models were either trained on nlp features only, eeg features only or a combination of both. All predictors were z-scored. An L2-logistic regression was chosen due to the small size of the feature spaces, and balanced accuracy was used to account for imbalance between target and nontarget trials.. The analysis included a comparison between performances of classifiers built on eeg features only, nlp features only and a combination of both. Within-subject cross-validation included 5-fold cross-validation for each subject separately, with classifiers trained on 4 and tested on 1. On the other hand, Across-subject cross-validation had the model trained on 29 subjects and tested on 1. In order to assess statistical significance a permutation test was performed, with 1000 permutations where labels were randomly shuffled. For each permutation the pipeline was repeated. The p-value was computed as the proportion of permuted mean accuracies exceeding the observed mean accuracy.

RESULTS

The NLP features correlation matrix showed that, except for surprisal-SyntacticComplexity and strokes-SyntacticComplexity pairs, the variables reached a significance with p<.001. Nevertheless, overall Pearson's r were all below 0.7.

Matrice di Correlazione

		Z_SyntacticComplexity	Z_Surprisal	Z_Strokes	Z_char_frequency	Z_word_frequency
Z_SyntacticComplexity	r di Pearson	—				
	gdl	—				
	valore p	—				
Z_Surprisal	r di Pearson	−0.033	—			
	gdl	11453	—			
	valore p	< .001	—			
Z_Strokes	r di Pearson	0.001	0.312	—		
	gdl	11453	11453	—		
	valore p	0.889	< .001	—		
Z_char_frequency	r di Pearson	−0.001	−0.297	−0.431	—	
	gdl	11453	11453	11453	—	
	valore p	0.933	< .001	< .001	—	
Z_word_frequency	r di Pearson	0.121	−0.277	−0.226	0.653	—
	gdl	11453	11453	11453	11453	—
	valore p	< .001	< .001	< .001	< .001	—

In order to strengthen such findings a linear regression model was carried out, with a continuous variable as VD and all NLP variables as independent variables: the VIF values were all below 3, therefore all NLP variables were kept in future analysis.

Statistiche di Collinearità

	VIF	Tolleranza
Z_SyntacticComplexity	1.03	0.974
Z_Surprisal	1.17	0.854
Z_Strokes	1.30	0.768
Z_char_frequency	2.08	0.482
Z_word_frequency	1.83	0.546

The independent samples t-test showed that the only feature showing significantly different means between groups was Z_Word_Frequency, t(274)=4.307, p<.001, d=0.5235.

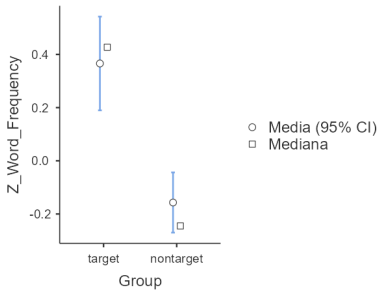
Independent Samples T-Test

		Statistic	df	p		Effect Size
Z_syntactic _complexity	Student's t	0.366	274	0.715	Cohen's d	0.0445
	Welch's t	0.370	263	0.712	Cohen's d	0.0447
Z_surprisal	Student's t	1.268	274	0.206	Cohen's d	0.1541
	Welch's t	1.258	246	0.210	Cohen's d	0.1535
Z_strokes	Student's t	1.172	274	0.242	Cohen's d	0.1425
	Welch's t	1.157	240	0.249	Cohen's d	0.1415
Z_char_frequency	Student's t	−1.001 [a]	274	0.318	Cohen's d	−0.1216
	Welch's t	−0.956	203	0.340	Cohen's d	−0.1188
Z_word_frequency	Student's t	4.307	274	< .001	Cohen's d	0.5235
	Welch's t	4.286	249	< .001	Cohen's d	0.5222

Note. H_a $\mu_{target} \neq \mu_{nontarget}$

[a] Levene's test is significant (p < .05), suggesting a violation of the assumption of equal variances

Z_Word_Frequency



A χ^2 test resulted significant $\chi^2(1)=72.9$, $p<.001$, meaning that between the target and nontarget groups the amount of repeated sentences was significantly not balanced.

		doubles		
Gruppo		DOUBLE	NOTDOUBLE	Totale
target	Osservato	1	119	120
	% di riga	0.8 %	99.2 %	100.0 %
nontarget	Osservato	123	157	280
	% di riga	43.9 %	56.1 %	100.0 %
Totale	Osservato	124	276	400
	% di riga	31.0 %	69.0 %	100.0 %

Test χ^2

	Valore	gdl	p
χ^2	72.9	1	<.001
N	400		

A paired t-test was carried out between the target and nontarget groups. The average amplitude in late Cz resulted as significantly higher in target than nontarget, $t(29)=-3.8$, $p<.001$, $d=-0.69$. The average amplitude in late C3 resulted as significantly different between target-nontarget, $t(29)=4.6$, $p<.001$, $d=-0.85$. The average alpha early global activation

resulted as significantly different between target-nontarget, t(29)=-2.561, p=.016, d=-0.4. The average beta early left activation resulted as significantly different between target-nontarget, t(29)=-3.4, p=.002, d=-0.63.

Paired Samples T-Test

			statistic	df	p		Effect Size
amp_early_chOz_Group0	amp_early_chOz_Group1	t di Student	−0.406	29.0	0.688	d di Cohen	−0.0742
amp_late_chCz_Group0	amp_late_chCz_Group1	t di Student	−3.810	29.0	< .001	d di Cohen	−0.6957
amp_late_chC3_Group0	amp_late_chC3_Group1	t di Student	4.681	29.0	< .001	d di Cohen	0.8547
amp_late_chPz_Group0	amp_late_chPz_Group1	t di Student	0.345	29.0	0.732	d di Cohen	0.0630
LN_alpha_late_global_target	LN_alpha_late_global_nontarget	Student's t	0.796	29.0	0.433	Cohen's d	0.1453
LN_alpha_early_global_target	LN_alpha_early_global_nontarget	Student's t	−2.561	29.0	0.016	Cohen's d	−0.4676
LN_beta_late_left_target	LN_beta_late_left_nontarget	Student's t	−0.337	29.0	0.738	Cohen's d	−0.0616
LN_beta_early_left_target	LN_beta_early_left_nontarget	Student's t	−3.708	29.0	< .001	Cohen's d	−0.6770

Note. H_a $\mu_{\text{Measure 1 - Measure 2}} \neq 0$

Descriptives

	N	Mean	Median	SD	SE
amp_early_chOz_Group0	30	0.6990	0.7091	0.655	0.1196
amp_early_chOz_Group1	30	0.7297	0.7159	0.540	0.0986
amp_late_chCz_Group0	30	−0.1283	−0.1392	0.188	0.0344
amp_late_chCz_Group1	30	−0.0579	−0.0444	0.158	0.0289
amp_late_chC3_Group0	30	−0.1489	−0.1326	0.281	0.0513
amp_late_chC3_Group1	30	−0.2533	−0.2173	0.245	0.0448
amp_late_chPz_Group0	30	−0.0467	−0.0626	0.158	0.0289
amp_late_chPz_Group1	30	−0.0535	−0.0183	0.155	0.0284
LN_alpha_late_global_target	30	−22.6155	−22.6328	0.645	0.1178
LN_alpha_late_global_nontarget	30	−22.6395	−22.6153	0.666	0.1217
LN_alpha_early_global_target	30	−22.7307	−22.7389	0.528	0.0964
LN_alpha_early_global_nontarget	30	−22.7049	−22.7088	0.509	0.0929
LN_beta_late_left_target	30	−24.1010	−24.1759	0.569	0.1039
LN_beta_late_left_nontarget	30	−24.0952	−24.1426	0.579	0.1057
LN_beta_early_left_target	30	−23.9203	−23.8693	0.594	0.1084
LN_beta_early_left_nontarget	30	−23.8910	−23.8641	0.590	0.1077

A Linear Mixed Model was built with Z_Late_C3 as dependent variables and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.088, $LRTX^2(20)=846.586$, p<.001, Marginal R^2 = 0.005, $LRTX^2(13)=34.559$, p<.001. No effects reached significance.

```
Z_Late_C3 ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal +
Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition +
Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes +
Group:Z_char_frequency + Group:Z_word_frequency + ( 1 + Group + repetition
| sub ) + ( 1 | label_doubles )
```

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.088	20	846.586	< .001
Marginal	0.005	13	34.559	< .001

[5]

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	3.7641	1	563	0.053
repetition	0.1439	1	488	0.705
Z_SyntacticComplexity	0.0155	1	383	0.901
Z_Surprisal	0.1042	1	354	0.747
Z_Strokes	0.7536	1	348	0.386
Z_char_frequency	0.7947	1	311	0.373
Z_word_frequency	3.4540	1	385	0.064
Group ✻ repetition	1.4074	1	704	0.236
Group ✻ Z_SyntacticComplexity	0.7277	1	383	0.394
Group ✻ Z_Surprisal	0.7597	1	354	0.384
Group ✻ Z_Strokes	0.1336	1	348	0.715
Group ✻ Z_char_frequency	1.7984	1	311	0.181
Group ✻ Z_word_frequency	2.7163	1	385	0.100

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	0.06142	0.0695	−0.07472	0.19756	84.3	0.884	0.379
Group1	nontarget - target	−0.18455	0.0951	−0.37102	0.00191	563.3	−1.940	0.053
repetition1	2 - 1	0.03719	0.0980	−0.15500	0.22937	487.9	0.379	0.705
Z_SyntacticComplexity	Z_SyntacticComplexity	0.00134	0.0107	−0.01973	0.02241	383.1	0.125	0.901
Z_Surprisal	Z_Surprisal	−0.00363	0.0112	−0.02566	0.01840	354.3	−0.323	0.747
Z_Strokes	Z_Strokes	−0.00984	0.0113	−0.03207	0.01238	347.5	−0.868	0.386
Z_char_frequency	Z_char_frequency	−0.01271	0.0143	−0.04066	0.01524	311.0	−0.891	0.373
Z_word_frequency	Z_word_frequency	0.02928	0.0158	−0.00160	0.06016	384.6	1.858	0.064
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	−0.21852	0.1842	−0.57957	0.14254	704.2	−1.186	0.236
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	−0.01834	0.0215	−0.06047	0.02380	383.1	−0.853	0.394
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	0.01959	0.0225	−0.02447	0.06365	354.2	0.872	0.384
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	−0.00829	0.0227	−0.05275	0.03617	347.5	−0.366	0.715
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	−0.03824	0.0285	−0.09413	0.01765	311.0	−1.341	0.181
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	0.05193	0.0315	−0.00983	0.11369	384.6	1.648	0.100

[6]

Random Components

Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00148	0.0384	0.00161
sub	(Intercept)	0.07974	0.2824	0.08008
	Group1	0.01157	0.1076	
	repetition1	0.03391	0.1842	
Residual		0.91599	0.9571	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−0.01252	−0.412
	(Intercept)	repetition1	0.00901	0.173
	Group1	repetition1	−0.01253	−0.633

A Linear Mixed Model was built with Z_Late_Cz as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.044, $LRTX^2(20)=369.136$, $p<.001$, Marginal R^2 = 0.003, $LRTX^2(13)=23.643$, $p=.035$. No effects reached significance.

```
Z_Late_Cz ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal +
Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition +
Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes +
Group:Z_char_frequency + Group:Z_word_frequency + ( 1 + Group + repetition
| sub ) + ( 1 | label_doubles )
```

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.044	20	369.136	< .001
Marginal	0.003	13	23.643	0.035

[5]

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	2.15968	1	592	0.142
repetition	0.02698	1	523	0.870
Z_SyntacticComplexity	0.00232	1	389	0.962
Z_Surprisal	3.21976	1	360	0.074
Z_Strokes	0.07810	1	353	0.780
Z_char_frequency	4.00317	1	316	0.046
Z_word_frequency	2.39351	1	390	0.123
Group ✻ repetition	0.47732	1	714	0.490
Group ✻ Z_SyntacticComplexity	1.54843	1	389	0.214
Group ✻ Z_Surprisal	0.48653	1	359	0.486
Group ✻ Z_Strokes	0.01291	1	353	0.910
Group ✻ Z_char_frequency	0.01464	1	316	0.904
Group ✻ Z_word_frequency	0.19606	1	390	0.658

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.03805	0.0571	−0.14995	0.07386	203	−0.6665	0.506
Group1	nontarget - target	0.14073	0.0958	−0.04698	0.32845	592	1.4696	0.142
repetition1	2 - 1	−0.01633	0.0994	−0.21121	0.17855	523	−0.1643	0.870
Z_SyntacticComplexity	Z_SyntacticComplexity	5.29e−4	0.0110	−0.02102	0.02208	389	0.0481	0.962
Z_Surprisal	Z_Surprisal	−0.02063	0.0115	−0.04316	0.00191	360	−1.7944	0.074
Z_Strokes	Z_Strokes	−0.00324	0.0116	−0.02598	0.01950	353	−0.2795	0.780
Z_char_frequency	Z_char_frequency	−0.02918	0.0146	−0.05777	−5.92e−4	316	−2.0008	0.046
Z_word_frequency	Z_word_frequency	0.02493	0.0161	−0.00666	0.05652	390	1.5471	0.123
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.13016	0.1884	−0.23913	0.49945	714	0.6909	0.490
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	0.02736	0.0220	−0.01574	0.07046	389	1.2444	0.214
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	0.01604	0.0230	−0.02903	0.06111	359	0.6975	0.486
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	0.00264	0.0232	−0.04284	0.04811	353	0.1136	0.910
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	−0.00353	0.0292	−0.06070	0.05365	316	−0.1210	0.904
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.01427	0.0322	−0.07744	0.04890	390	−0.4428	0.658

[6]

Random Components

Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00155	0.0394	0.00161
sub	(Intercept)	0.02980	0.1726	0.03016
	Group1	0.00323	0.0568	
	repetition1	0.03031	0.1741	
Residual		0.95828	0.9789	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−6.86e−4	−0.0699
	(Intercept)	repetition1	−0.00650	−0.2161
	Group1	repetition1	−0.00712	−0.7191

A Linear Mixed Model was built with Z:Alpha_Early_Global as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.629, LRTX^2(20)=10796.971, p<.001, Marginal R^2 = 0.002, LRTX^2(13)=32.938, p=.002.
The interaction effect Group*char_frequency reached significance controlling for the other independent variables, F(1,298)=5.8245, p=.016.

Z_Alpha_Early_Global ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal + Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition + Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes + Group:Z_char_frequency + Group:Z_word_frequency + (1 + repetition + Group | sub) + (1 | label_doubles)

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.629	20	10796.971	< .001
Marginal	0.002	13	32.938	0.002

[5]

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	1.4978	1	671	0.221
repetition	0.0303	1	390	0.862
Z_SyntacticComplexity	1.9112	1	353	0.168
Z_Surprisal	0.6291	1	329	0.428
Z_Strokes	0.0454	1	326	0.831
Z_char_frequency	0.1736	1	298	0.677
Z_word_frequency	0.1084	1	354	0.742
Group ✻ repetition	1.0096	1	1027	0.315
Group ✻ Z_SyntacticComplexity	2.1160	1	353	0.147
Group ✻ Z_Surprisal	0.3773	1	329	0.539
Group ✻ Z_Strokes	0.4031	1	326	0.526
Group ✻ Z_char_frequency	5.8245	1	298	0.016
Group ✻ Z_word_frequency	1.3909	1	354	0.239

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.02280	0.15278	−0.32229	0.27668	31.5	−0.149	0.882
Group1	nontarget - target	0.08141	0.06652	−0.04898	0.21181	670.6	1.224	0.221
repetition1	2 - 1	0.01218	0.07002	−0.12507	0.14943	389.7	0.174	0.862
Z_SyntacticComplexity	Z_SyntacticComplexity	0.01042	0.00754	−0.00435	0.02519	353.4	1.382	0.168
Z_Surprisal	Z_Surprisal	0.00627	0.00791	−0.00923	0.02177	329.3	0.793	0.428
Z_Strokes	Z_Strokes	0.00170	0.00798	−0.01395	0.01735	326.1	0.213	0.831
Z_char_frequency	Z_char_frequency	−0.00420	0.01008	−0.02397	0.01556	297.8	−0.417	0.677
Z_word_frequency	Z_word_frequency	0.00364	0.01104	−0.01801	0.02529	354.1	0.329	0.742
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.12598	0.12538	−0.11978	0.37174	1026.6	1.005	0.315
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	−0.02193	0.01507	−0.05147	0.00762	353.4	−1.455	0.147
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	0.00971	0.01581	−0.02128	0.04071	329.2	0.614	0.539
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	−0.01014	0.01597	−0.04143	0.02116	326.2	−0.635	0.526
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.04867	0.02017	0.00914	0.08820	297.8	2.413	0.016
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.02605	0.02209	−0.06935	0.01725	354.0	−1.179	0.239

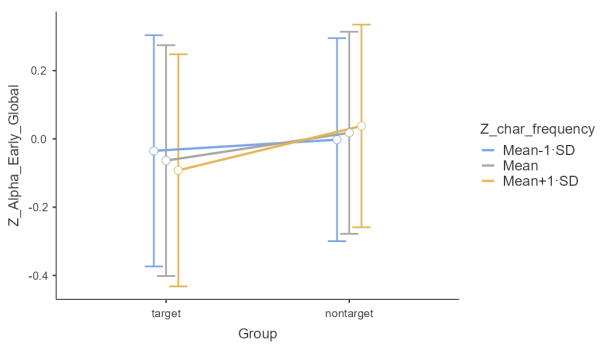
[6]

Simple effects showed that only when the mean frequency of the sentence, computed based on character frequencies, is 1 SD over the mean there is a trend towards group effect on the global alpha early activation, that is a trend towards a difference between target and nontarget. F(1,643)=3.446, p=.064.

ANOVA for Simple Effects of Group

Moderator Z_char_frequency	F	Num df	Den df	p
Mean-1·SD	0.226	1.00	648	0.635
Mean	1.498	1.00	671	0.221
Mean+1·SD	3.446	1.00	643	0.064

Plot: Z_Alpha_Early_Global ~ Group ✻ Z_char_frequency



A Linear Mixed Model was built with Z_beta_early_left as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.687, $LRTX^2(17)=112724.414$, p<.001, Marginal R^2 = 0.003, $LRTX^2(13)=34.696$, p<.001. The model originally had Group|Sub as a random slope, but it failed to converge, therefore it was excluded because of its small variance.
Syntactic Complexity reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=0.1626, IC=0.00294-0.00296, t(353.8)=2.392, p=.017. When syntactic complexity increases by one unit, beta early left activation significantly increases by 0.1626.

Moreover char_frequency reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=-0.01876, IC=0.03656–9.71e-4, t(295.8)=-2.067, p=.040. When sentence frequency based on character frequency increases by one unit, beta early left activation significantly decreases by 0.01876.

Z_beta_early_left ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal + Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition + Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes + Group:Z_char_frequency + Group:Z_word_frequency + (1 + repetition | sub) + (1 | label_doubles)

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.687	17	12724.414	< .001
Marginal	0.003	13	34.696	< .001

[5]

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	3.2199	1	801	0.073
repetition	0.0205	1	453	0.886
Z_SyntacticComplexity	5.7226	1	354	0.017
Z_Surprisal	1.4683	1	329	0.226
Z_Strokes	0.0597	1	325	0.807
Z_char_frequency	4.2733	1	296	0.040
Z_word_frequency	0.8142	1	355	0.367
Group ✻ repetition	2.8326	1	930	0.093
Group ✻ Z_SyntacticComplexity	1.4604	1	354	0.228
Group ✻ Z_Surprisal	1.2254	1	329	0.269
Group ✻ Z_Strokes	0.0148	1	325	0.903
Group ✻ Z_char_frequency	2.1177	1	296	0.147
Group ✻ Z_word_frequency	1.3381	1	355	0.248

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.03035	0.15530	−0.33476	0.2741	31.0	−0.195	0.846
Group1	nontarget - target	0.10365	0.05776	−0.00957	0.2169	800.7	1.794	0.073
repetition1	2 - 1	0.00896	0.06263	−0.11380	0.1317	453.3	0.143	0.886
Z_SyntacticComplexity	Z_SyntacticComplexity	0.01626	0.00680	0.00294	0.0296	353.8	2.392	0.017
Z_Surprisal	Z_Surprisal	0.00863	0.00713	−0.00533	0.0226	329.0	1.212	0.226
Z_Strokes	Z_Strokes	−0.00176	0.00719	−0.01586	0.0123	325.4	−0.244	0.807
Z_char_frequency	Z_char_frequency	−0.01876	0.00908	−0.03656	−9.71e−4	295.8	−2.067	0.040
Z_word_frequency	Z_word_frequency	0.00899	0.00996	−0.01054	0.0285	354.6	0.902	0.367
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.19176	0.11394	−0.03158	0.4151	930.4	1.683	0.093
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	−0.01643	0.01359	−0.04307	0.0102	353.8	−1.208	0.228
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	−0.01577	0.01425	−0.04371	0.0122	329.0	−1.107	0.269
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	−0.00175	0.01439	−0.02995	0.0264	325.4	−0.122	0.903
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.02642	0.01815	−0.00917	0.0620	295.8	1.455	0.147
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.02304	0.01992	−0.06209	0.0160	354.6	−1.157	0.248

Random Components

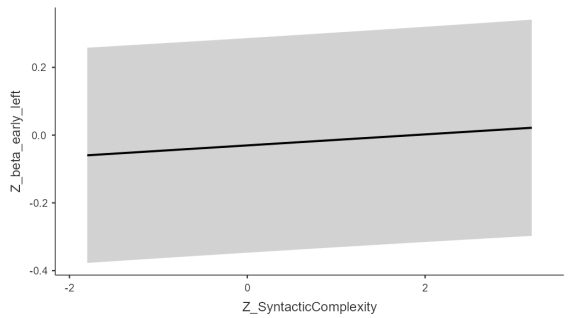
Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00190	0.0436	0.00589
sub	(Intercept)	0.69849	0.8358	0.68532
	repetition1	0.02031	0.1425	
Residual		0.32072	0.5663	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

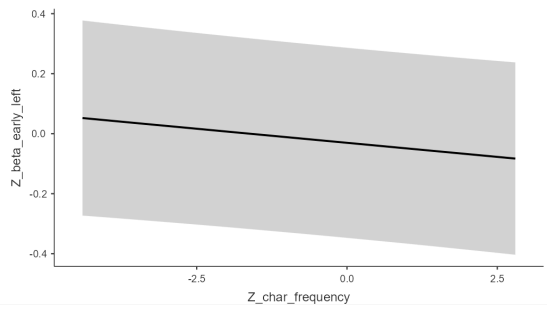
Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	repetition1	0.0143	0.120

Plot: Z_beta_early_left ~ Z_SyntacticComplexity



Plot: Z_beta_early_left ~ Z_char_frequency



A Linear Mixed Model was built with Z_Early_Oz as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.041, LRTX^2(20)=315.246, p<.001, Marginal R^2 = 0.001, LRTX^2(13)=6.383, p=.931. No effects reached significance.

Z_Early_Oz ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal + Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition + Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes + Group:Z_char_frequency + Group:Z_word_frequency + (1 + Group + repetition | sub) + (1 | label_doubles)

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.041	20	315.246	< .001
Marginal	0.001	13	6.383	0.931

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	0.0385	1	658	0.844
repetition	0.3401	1	689	0.560
Z_SyntacticComplexity	1.6484	1	406	0.200
Z_Surprisal	1.0306	1	377	0.311
Z_Strokes	0.5391	1	371	0.463
Z_char_frequency	0.6955	1	335	0.405
Z_word_frequency	2.7351	1	407	0.099
Group ✻ repetition	0.2502	1	879	0.617
Group ✻ Z_SyntacticComplexity	0.6615	1	406	0.417
Group ✻ Z_Surprisal	0.4783	1	377	0.490
Group ✻ Z_Strokes	0.2320	1	371	0.630
Group ✻ Z_char_frequency	0.1825	1	335	0.670
Group ✻ Z_word_frequency	1.9170	1	407	0.167

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	0.01444	0.0597	−0.10258	0.13146	187	0.242	0.809
Group1	nontarget - target	−0.01969	0.1003	−0.21632	0.17693	658	−0.196	0.844
repetition1	2 - 1	0.05819	0.0998	−0.13741	0.25379	689	0.583	0.560
Z_SyntacticComplexity	Z_SyntacticComplexity	−0.01457	0.0113	−0.03681	0.00767	406	−1.284	0.200
Z_Surprisal	Z_Surprisal	0.01206	0.0119	−0.01123	0.03535	377	1.015	0.311
Z_Strokes	Z_Strokes	−0.00880	0.0120	−0.03231	0.01470	371	−0.734	0.463
Z_char_frequency	Z_char_frequency	−0.01260	0.0151	−0.04220	0.01701	335	−0.834	0.405
Z_word_frequency	Z_word_frequency	0.02751	0.0166	−0.00510	0.06011	407	1.654	0.099
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	−0.09641	0.1927	−0.47421	0.28139	879	−0.500	0.617
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	0.01846	0.0227	−0.02603	0.06295	406	0.813	0.417
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	−0.01643	0.0238	−0.06301	0.03014	377	−0.692	0.490
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	0.01155	0.0240	−0.03546	0.05856	371	0.482	0.630
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.01290	0.0302	−0.04630	0.07211	335	0.427	0.670
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.04606	0.0333	−0.11126	0.01915	407	−1.385	0.167

Random Components

Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00338	0.0581	0.00351
sub	(Intercept)	0.03556	0.1886	0.03569
	Group1	0.01646	0.1283	
	repetition1	0.02010	0.1418	
Residual		0.96086	0.9802	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−0.00944	−0.390
	(Intercept)	repetition1	0.00505	0.189
	Group1	repetition1	−0.01424	−0.783

A Linear Mixed Model was built with Z_Late_Pz as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.033, $LRTX^2(20)=264.131$, p<.001, Marginal R^2 = 0.003, $LRTX^2(13)=29.297$, p=.006.

Surprisal reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=-0.04166, IC=-0.0638- -0.01950, t(393)=-3.6851, p=.017. When surprisal increases by one unit, Pz late amplitude significantly decreases by 0.04166.

Moreover char_frequency reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=-0.03205, IC=-0.0601– -0.00398, t(342)=-2.2379, p=.026. When sentence frequency based on character frequency increases by one unit,Pz late amplitude significantly decreases by 0.03205.

The interaction effect Group*strokes reached significance controlling for the other independent variables, F(1,384)=8.38103, p=.004.

```
Z_Late_Pz ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal +
Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition +
Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes +
Group:Z_char_frequency + Group:Z_word_frequency + ( 1 + Group + repetition
| sub ) + ( 1 | label_doubles )
```

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.033	20	264.131	< .001
Marginal	0.003	13	29.297	0.006

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	0.39384	1	570	0.531
repetition	0.78746	1	587	0.375
Z_SyntacticComplexity	0.04396	1	426	0.834
Z_Surprisal	13.58016	1	393	< .001
Z_Strokes	1.02400	1	384	0.312
Z_char_frequency	5.00804	1	342	0.026
Z_word_frequency	0.49048	1	428	0.484
Group ✻ repetition	0.83669	1	692	0.361
Group ✻ Z_SyntacticComplexity	0.00267	1	426	0.959
Group ✻ Z_Surprisal	1.46053	1	393	0.228
Group ✻ Z_Strokes	8.38103	1	384	0.004
Group ✻ Z_char_frequency	0.00897	1	342	0.925
Group ✻ Z_word_frequency	1.68026	1	427	0.196

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.02971	0.0559	−0.1394	0.07994	211	−0.5312	0.596
Group1	nontarget - target	0.05977	0.0952	−0.1269	0.24644	570	0.6276	0.531
repetition1	2 - 1	−0.08511	0.0959	−0.2731	0.10290	587	−0.8874	0.375
Z_SyntacticComplexity	Z_SyntacticComplexity	0.00227	0.0108	−0.0189	0.02348	426	0.2097	0.834
Z_Surprisal	Z_Surprisal	−0.04166	0.0113	−0.0638	−0.01950	393	−3.6851	< .001
Z_Strokes	Z_Strokes	−0.01154	0.0114	−0.0339	0.01081	384	−1.0119	0.312
Z_char_frequency	Z_char_frequency	−0.03205	0.0143	−0.0601	−0.00398	342	−2.2379	0.026
Z_word_frequency	Z_word_frequency	−0.01111	0.0159	−0.0422	0.01998	428	−0.7003	0.484
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.17051	0.1864	−0.1949	0.53590	692	0.9147	0.361
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	−0.00112	0.0216	−0.0435	0.04130	426	−0.0517	0.959
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	0.02732	0.0226	−0.0170	0.07164	393	1.2085	0.228
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	0.06603	0.0228	0.0213	0.11074	384	2.8950	0.004
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.00271	0.0286	−0.0534	0.05885	342	0.0947	0.925
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	0.04112	0.0317	−0.0211	0.10330	427	1.2962	0.196

Random Components

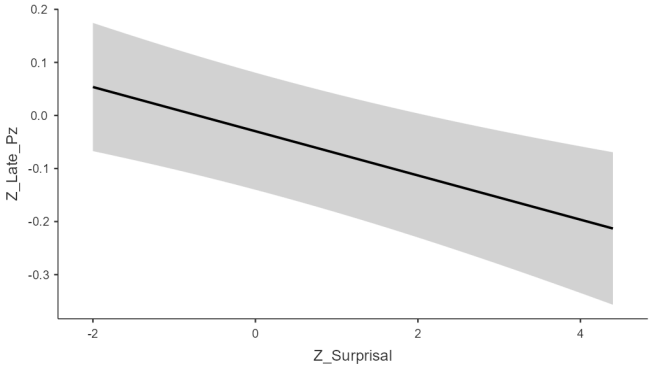
Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	3.28e−4	0.0181	3.38e−4
sub	(Intercept)	0.02746	0.1657	0.0276
	Group1	0.00646	0.0804	
	repetition1	0.01537	0.1240	
Residual		0.96929	0.9845	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

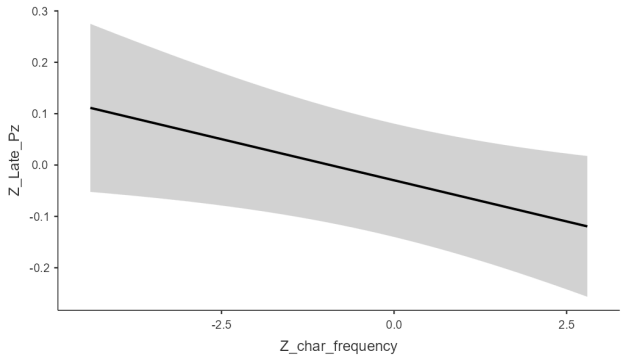
Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−4.84e−4	−0.0364
	(Intercept)	repetition1	0.00554	0.2699
	Group1	repetition1	−0.00630	−0.6317

Plot: Z_Late_Pz ~ Z_Surprisal



Plot: Z_Late_Pz ~ Z_char_frequency

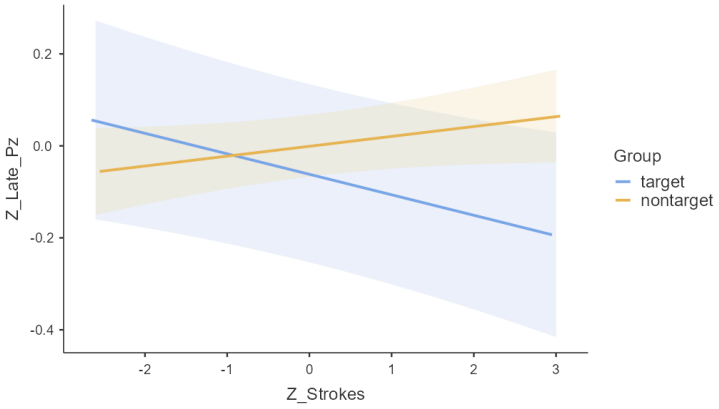


Simple effects showed that only for the target group there is a strokes effect on the Pz late amplitude, b = -0.0446, IC=0.0187- -0.08128, t(615)=-2.38, p=.017, meaning that when strokes increases by 1 unit, Pz late amplitude decreases by 0.0446.

Parameter Estimates for simple effects of Z_Strokes

Moderator				95% Confidence Intervals				
Group	Effect	Estimate	SE	Lower	Upper	df	t	p
target	Z_Strokes	−0.0446	0.0187	−0.08128	−0.00783	615	−2.38	0.017
nontarget	Z_Strokes	0.0215	0.0131	−0.00429	0.04724	187	1.64	0.102

Plot: Z_Late_Pz ~ Z_Strokes ✻ Group



A Linear Mixed Model was built with alpha late global as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.564, LRTX^2(20)=88818.207, p<.001, Marginal R^2 = 0.003, LRTX^2(13)=21.454, p=.064.

Surprisal reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=0.02712, IC=0.00763-0.0466, t(295)=2.7275, p=.007. When surprisal increases by one unit, alpha late global activation significantly increases by 0.02712.

Z_Alpha_Late_Global ~ 1 + Group + repetition + Z_SyntacticComplexity + Z_Surprisal + Z_Strokes + Z_char_frequency + Z_word_frequency + Group:repetition + Group:Z_SyntacticComplexity + Group:Z_Surprisal + Group:Z_Strokes + Group:Z_char_frequency + Group:Z_word_frequency + (1 + Group + repetition | sub) + (1 | label_doubles)

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.564	20	8818.207	< .001
Marginal	0.003	13	21.454	0.064

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	0.336	1	321	0.563
repetition	1.18e−4	1	315	0.991
Z_SyntacticComplexity	2.846	1	295	0.093
Z_Surprisal	7.439	1	278	0.007
Z_Strokes	0.827	1	278	0.364
Z_char_frequency	0.651	1	261	0.420
Z_word_frequency	0.181	1	295	0.671
Group ✻ repetition	2.098	1	1726	0.148
Group ✻ Z_SyntacticComplexity	0.481	1	295	0.488
Group ✻ Z_Surprisal	0.154	1	278	0.695
Group ✻ Z_Strokes	0.990	1	278	0.321
Group ✻ Z_char_frequency	0.746	1	261	0.389
Group ✻ Z_word_frequency	1.092	1	295	0.297

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.02367	0.14706	−0.31193	0.2646	32.9	−0.1609	0.873
Group1	nontarget - target	0.04982	0.08597	−0.11870	0.2183	321.5	0.5795	0.563
repetition1	2 - 1	−9.18e−4	0.08445	−0.16645	0.1646	315.4	−0.0109	0.991
Z_SyntacticComplexity	Z_SyntacticComplexity	0.01591	0.00943	−0.00258	0.0344	295.0	1.6869	0.093
Z_Surprisal	Z_Surprisal	0.02712	0.00994	0.00763	0.0466	278.3	2.7275	0.007
Z_Strokes	Z_Strokes	0.00913	0.01004	−0.01055	0.0288	278.2	0.9093	0.364
Z_char_frequency	Z_char_frequency	0.01030	0.01276	−0.01472	0.0353	260.6	0.8070	0.420
Z_word_frequency	Z_word_frequency	0.00588	0.01382	−0.02122	0.0330	294.9	0.4252	0.671
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.21098	0.14565	−0.07452	0.4965	1726.0	1.4485	0.148
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	−0.01309	0.01887	−0.05007	0.0239	295.0	−0.6936	0.488
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	−0.00781	0.01989	−0.04680	0.0312	278.3	−0.3929	0.695
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	−0.01999	0.02009	−0.05936	0.0194	278.2	−0.9951	0.321
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.02204	0.02552	−0.02798	0.0721	260.6	0.8637	0.389
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.02889	0.02765	−0.08309	0.0253	294.9	−1.0449	0.297

Random Components

Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00868	0.0931	0.0191
sub	(Intercept)	0.60718	0.7792	0.5770
	Group1	0.05527	0.2351	
	repetition1	0.05482	0.2341	
Residual		0.44516	0.6672	

Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−0.0874	−0.477
	(Intercept)	repetition1	0.0827	0.453
	Group1	repetition1	−0.0316	−0.574

A Linear Mixed Model was built with beta late left as a dependent variable and Group, repetition and NLP features as independent variables. The model reached significance, Conditional R^2 = 0.756, LRTX^2(20)=15401.312, p<.001, Marginal R^2 = 0.001, LRTX^2(13)=10.396, p=.661.

Word_frequency reached significance controlling for the other independent variables, that is it has a significant effect on beta early left activation, b=0.02135, IC=0.00283-0.03986, t(342.6)=2.2602, p=.024. When word_frequency increases by one unit, beta late left activation significantly increases by 0.02135.

```
Z_beta_late_left ~ 1 + Group + repetition + Z_SyntacticComplexity +
Z_Surprisal + Z_Strokes + Z_char_frequency + Z_word_frequency +
Group:repetition + Group:Z_SyntacticComplexity + Group:Z_Surprisal +
Group:Z_Strokes + Group:Z_char_frequency + Group:Z_word_frequency + ( 1 +
Group + repetition | sub ) + ( 1 | label_doubles )
```

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.756	20	15401.312	< .001
Marginal	0.001	13	10.396	0.661

Fixed Effects Omnibus Tests

	F	df	df (res)	p
Group	0.3261	1	706	0.568
repetition	0.3079	1	160	0.580
Z_SyntacticComplexity	0.0268	1	342	0.870
Z_Surprisal	0.3292	1	320	0.567
Z_Strokes	0.0103	1	319	0.919
Z_char_frequency	1.1375	1	294	0.287
Z_word_frequency	5.1085	1	343	0.024
Group ✻ repetition	0.3108	1	1268	0.577
Group ✻ Z_SyntacticComplexity	0.1270	1	342	0.722
Group ✻ Z_Surprisal	0.2605	1	320	0.610
Group ✻ Z_Strokes	0.1072	1	319	0.744
Group ✻ Z_char_frequency	2.5351	1	294	0.112
Group ✻ Z_word_frequency	3.3383	1	343	0.069

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	−0.00436	0.16799	−0.33365	0.32494	30.4	−0.0259	0.979
Group1	nontarget - target	0.03205	0.05612	−0.07796	0.14206	706.4	0.5711	0.568
repetition1	2 - 1	0.03716	0.06698	−0.09412	0.16845	159.8	0.5549	0.580
Z_SyntacticComplexity	Z_SyntacticComplexity	0.00106	0.00644	−0.01158	0.01369	342.3	0.1638	0.870
Z_Surprisal	Z_Surprisal	0.00389	0.00677	−0.00939	0.01716	320.2	0.5737	0.567
Z_Strokes	Z_Strokes	−6.93e−4	0.00684	−0.01410	0.01271	318.5	−0.1013	0.919
Z_char_frequency	Z_char_frequency	−0.00923	0.00866	−0.02620	0.00774	293.7	−1.0665	0.287
Z_word_frequency	Z_word_frequency	0.02135	0.00944	0.00283	0.03986	342.6	2.2602	0.024
Group1 ✻ repetition1	(nontarget - target) ✻ (2 - 1)	0.05845	0.10484	−0.14706	0.26397	1267.6	0.5575	0.577
Group1 ✻ Z_SyntacticComplexity	(nontarget - target) ✻ Z_SyntacticComplexity	0.00459	0.01289	−0.02067	0.02986	342.3	0.3564	0.722
Group1 ✻ Z_Surprisal	(nontarget - target) ✻ Z_Surprisal	−0.00691	0.01354	−0.03346	0.01964	320.2	−0.5104	0.610
Group1 ✻ Z_Strokes	(nontarget - target) ✻ Z_Strokes	0.00448	0.01368	−0.02233	0.03129	318.5	0.3274	0.744
Group1 ✻ Z_char_frequency	(nontarget - target) ✻ Z_char_frequency	0.02757	0.01732	−0.00637	0.06151	293.7	1.5922	0.112
Group1 ✻ Z_word_frequency	(nontarget - target) ✻ Z_word_frequency	−0.03451	0.01889	−0.07153	0.00251	342.6	−1.8271	0.069

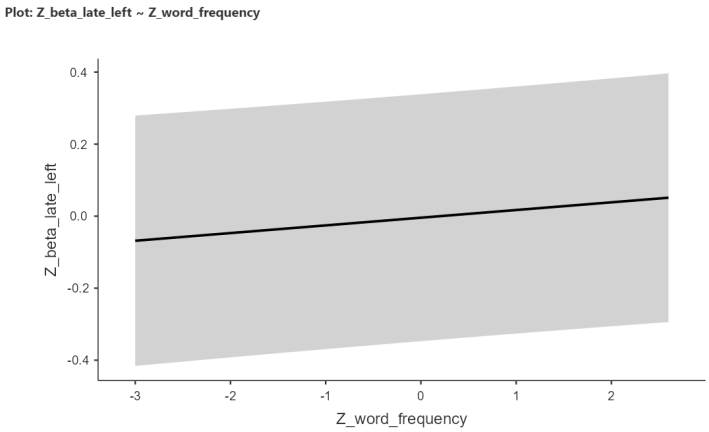
Random Components

Groups	Name	Variance	SD	ICC
label_doubles	(Intercept)	0.00277	0.0527	0.0109
sub	(Intercept)	0.82531	0.9085	0.7664
	Group1	0.00915	0.0956	
	repetition1	0.05213	0.2283	
Residual		0.25158	0.5016	

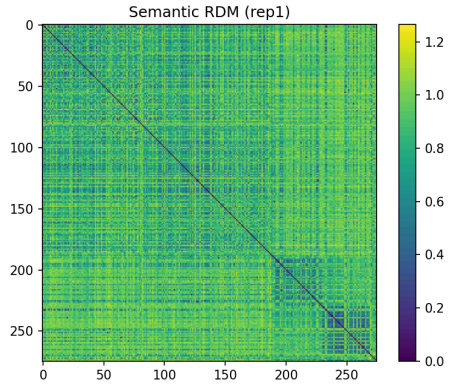
Note. Number of Obs: 11455 , Number of groups: label_doubles 275, sub 30

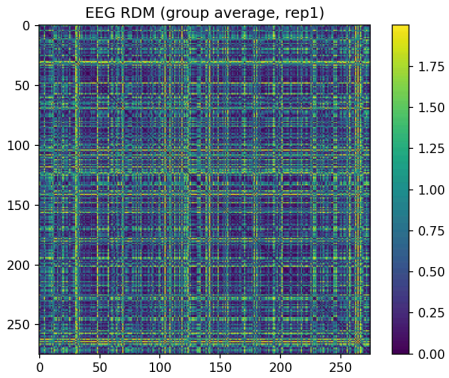
Random Parameters correlations

Groups	param1	param2	Cov.	Corr.
sub	(Intercept)	Group1	−0.02272	−0.262
	(Intercept)	repetition1	0.13966	0.673
	Group1	repetition1	−0.00617	−0.282



Using the same eeg features, first the correlation between a RDM based only on first sentence occurrences and a RDM based only on second occurrences had an output of a mean rho across subjects of ~ 0.007. Therefore, using only first occurrences a RSA was carried out between the semantic RSM and eeg RSM across all subjects, but it did not reach significance following a permutation test, rho(mean)=0.007, p-value: 0.1055, d= 0.30.





Regarding within subject classification, permutation testing indicated that all models performed significantly (p<.001), NLP-only model accuracy = 0.65, EEG-only model accuracy = 0.53, NLP+EEG model accuracy = 0.65. On the other hand, across subject

classification models also all performed significantly ($p<.001$), with NLP-only model accuracy = 0.68, EEG-only model accuracy = 0.53, NLP+EEG model accuracy = 0.68.

DISCUSSION

The paired t-test, carried out between the target and nontarget groups to replicate the original result, only showed some of the expected results, possibly due to methodological differences in preprocessing pipelines: only the amplitude in late Cz and late C3 resulted as significantly higher in target than nontarget; moreover only the alpha early global activation and the beta early left activation resulted as significantly different between target-nontarget.

The NLP features in the current study did not exhibit problematic collinearity as, although the variables reached significance in the correlation matrix, all Pearson's r were below 0.7 and VIF values were below 3. This indicates that each feature provides unique information, allowing the model to clearly isolate the individual effect of each.

The sentences used in TMNRED were originally balanced for length, strokes count and word frequency [1]. Nevertheless, word frequency, as computed in the current study, was significantly different between target and nontarget groups, while all other features showed no difference as expected.

Moreover, the amount of repeated sentences was significantly not balanced between target and nontarget sentences, adding validity and foundation to the mixed model approach used in the current study.

The Linear Mixed Model, accounting for repetitions, linguistic features, random subject and sentence variability, showed that, regarding channel components, early Oz, late C3 and late Cz amplitudes had low explained variance by both fixed and random effects and fixed effects only, and no significant effects.

Regarding the RSA, the repetition of sentences posed a serious issue as RSA needed unique trials, therefore, in order to decide whether repetitions could be averaged, a correlation between the RDM based on the first occurrence of repeated sentences and RDM based on the second occurrence was carried out. The results suggested that second occurrences and first occurrences could not be interpreted as equivalent responses differentiated by just noise, therefore a deletion-based approach, that is leaving the second occurrence trials out of the RSA. The RSA results showed that the embedding-based semantic similarity was not aligned with the neural representational structure extracted from EEG features.

These results showed that group effects on EEG features are attenuated when controlling for repetitions, linguistic features, random subject and sentence variability. Moreover, the RSA result could further support semantic-related EEG features having weak direct relation with semantic aspects of the sentences.

In contrast, Late Pz also had low explained variance by both fixed and random effects and fixed effects only, but Surprisal had a significant effect, that is the more surprisal increased the more Pz late amplitude decreased; in addition, sentence frequency, derived from characters' frequencies, had also a significant effect, meaning that the more sentence frequency increased the more Pz late amplitude decreased. An interaction effect Group*strokes also reached significance, which could be interpreted as only for the target

group there is a strokes effect on the PZ late amplitude, that is the higher average number of strokes is the more the late PZ amplitude decreases. While for high frequency sentences a decrease is expected, such evidence represents an unexpected result for surprising or high-strokes-number stimuli, which are typically associated with large amplitudes [9, 10]. In this sense, a possible insight could be derived from single stream models of N400 and P600 posterior neural responses. These models posit that N400 underlies lexical retrieval and P600 reflects the integration of retrieved word meaning, the former lingers until 600 ms while the latter emerges around 600 ms, meaning that within the 500 ms window used in the current study an overlap between the two components could elicit such result in posterior channels when exposed to surprising stimuli[9]. Moreover, beta early left activation had low explained variance by fixed effects only and high explained variance by both random and fixed effects, but syntactic complexity had a significant effect while controlling for other independent variables, that is the more syntactic complexity increased the more beta early left activation significantly increased, and moreover sentence frequency, derived from characters' frequencies, had also a significant effect, meaning that the more sentence frequency increased the more beta early left activation decreased. Alpha late global activation also showed low explained variance by fixed effects only and high explained variance by both random and fixed effects, with Surprisal having a significant effect while controlling for the other independent variables, that is the more surprisal increased the more alpha late global activation increased. Finally, beta late left activation, also showed low explained variance by fixed effects only and high explained variance by both random and fixed effects, with sentence frequency derived from words's frequencies having a significant effect, meaning that the more sentence frequency increased the more beta late left activation increased. Overall, all mixed models with alpha and beta activity as dependent variables had marginal R^2 and substantially higher conditional R^2, meaning that a large proportion of variance is attributable to between-subject and between-trials variability. The fixed effects of nlp features on alpha and beta are consistent with the increased activation of these brain rhythms related to highly cognitive loaded processes, in particular semantic processes [8].

Target-nontarget classification was primarily driven by linguistic features, with balanced accuracy above chance, whereas EEG features alone only yielded marginally above-chance decoding accuracy in both cases. Combining EEG and linguistic features did not improve performance compared to linguistic features alone, in both within and across subject cases. These results suggest that the discriminative structure between target-nontarget is captured mainly by the linguistic properties of the stimuli, aligning with mixed models results where group effects were attenuated when controlling for linguistic predictors, and with RSA findings of weak alignment between neural and semantic representational spaces.

In conclusion, when controlling for repetitions, linguistic features and hierarchical variability, the experimental group effect on eeg features is attenuated. Moreover, eeg semantic-related features seem not to reflect semantic similarity between sentences and neither to be driving the classification into target or nontarget groups. Linguistic features and between-subject and between trial variability seem to explain more variance than grouping, suggesting that stimulus-level structure accounts for most of the differentiation between groups.

Limitations

Some of the theoretical interpretations draws on neurolinguistics studies on alphabetical languages such as Dutch and English. Epoching and time-window selection were constrained by the available event markers in the dataset, which only marked the start of the sentence. Finally, the relatively broad time windows may include possible overlaps of ERP components, as possibly suggested by the pattern observed for the late Pz.

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